

SIXTH FORM DISCRETE

Mock Mechanics Paper

Time Allowed: 1 hour 30 minutes • Total Marks: 75

Candidate Name _____

Candidate Number _____

Instructions

- Use black or dark blue pen. Pencil may be used for diagrams.
- Answer **all** questions.
- Unless otherwise stated, take $g = 9.8 \text{ m s}^{-2}$ and give answers to an appropriate degree of accuracy.
- Non-exact numerical answers should be given to 3 significant figures unless stated otherwise.
- The total number of marks for this paper is 75.
- You are permitted to use a scientific calculator.

1. A car is travelling along a straight horizontal road. At time $t = 0$ the car passes a point A with speed 25 m s^{-1} . The car accelerates uniformly for 8 s until it reaches a speed of 45 m s^{-1} . It then travels at this constant speed for a further 20 s before applying the brakes and decelerating uniformly to rest at point B.
[10 marks]

- (a) Sketch a velocity-time graph for the motion of the car from A to B, clearly labelling the times and velocities at each stage. [2]
- (b) Find the total time taken for the car to travel from A to B. [2]
- (c) Find the total distance travelled from A to B. [3]
- (d) Find the deceleration of the car during the final stage of the motion, given that the distance travelled during deceleration is 180 m. [3]

2. A particle P of mass 4 kg lies on a rough horizontal table and is attached to one end of a light inextensible string which passes over a small smooth pulley fixed at the edge of the table. The other end of the string is attached to a particle Q of mass 3 kg which hangs vertically below the pulley. The system is released from rest with the string taut. The coefficient of friction between particle P and the table is μ . The tension in the string is 24.5 N. [8 marks]

- (a) By considering particle Q, show that the acceleration of the system is 1.63 m s^{-2} (to 3 s.f.). [3]
- (b) Find the value of μ . [5]

3. A lift of mass 500 kg is descending vertically. The lift is supported by a light vertical cable. Inside the lift, a box of mass 40 kg rests on the floor. [12 marks]

- (a) Draw separate force diagrams showing all the forces acting on (i) the lift, and (ii) the box. [3]
- (b) Given that the tension in the cable is 4200 N, find the magnitude of the acceleration of the lift. [3]
- (c) Hence find the magnitude of the force exerted by the floor of the lift on the box. [3]
- (d) A person of mass 70 kg now also stands in the lift on top of the box. By considering the system as a whole, find the new tension in the cable needed to produce the same downward acceleration found in part (b). [3]

4. A particle of mass 6 kg is projected with speed 18 m s^{-1} directly up a rough plane inclined at 30° to the horizontal. The coefficient of friction between the particle and the plane is 0.25. [10 marks]

- (a) Find the magnitude of the frictional force acting on the particle. [3]
- (b) By resolving parallel to the plane, find the deceleration of the particle. [3]
- (c) Show that the particle comes to rest after travelling approximately 14.7 m up the plane. [3]
- (d) State, with a reason, whether the particle will begin to slide back down the plane once it comes to rest. [1]

5. Two spheres A and B of masses 5 kg and 3 kg respectively are moving towards each other along the same straight line on a smooth horizontal surface. Sphere A has speed 4 m s^{-1} and sphere B has speed 2 m s^{-1} . After the collision sphere B moves in the same direction as A was originally moving, with speed 3 m s^{-1} .

[11 marks]

- (a)** Find the speed and direction of sphere A after the collision. [4]
- (b)** Find the coefficient of restitution e for the collision. [3]
- (c)** Calculate the loss of kinetic energy in the collision. [4]

6. A particle is projected horizontally from the top of a vertical cliff of height 78.4 m above the sea with a speed of 24.5 m s^{-1} . [11 marks]

- (a)** Find the time taken for the particle to reach the sea. [2]
- (b)** Find the horizontal distance from the base of the cliff to the point where the particle hits the sea. [2]
- (c)** Find the speed and direction of motion of the particle when it reaches the sea. [4]
- (d)** A second particle is projected from the same point at the same speed 24.5 m s^{-1} , but at an angle of 30° above the horizontal. Find the horizontal distance travelled by this particle before it hits the sea. [3]

7. A body of mass 8 kg moves in a straight line across a rough horizontal surface. At point A the speed of the body is 12 m s^{-1} . A constant horizontal force of magnitude 30 N acts on the body in the direction of motion. As the body moves from A to B, a distance of 20 m, the speed of the body increases to 14 m s^{-1} . [13 marks]

- (a)** Using the work-energy principle, or otherwise, find the magnitude of the work done against the resistances to motion as the body moves from A to B. [4]
- (b)** Hence find the magnitude of the resistive force, assuming it is constant. [2]

A second body of mass 12 kg is projected from B with speed 5 m s^{-1} up a smooth plane inclined at an angle θ to the horizontal, where $\sin \theta = 1/7$. The same constant force of 30 N now acts on the body, but directed **down** the plane.

- (c)** Find the acceleration of the body. [4]
- (d)** The body comes to instantaneous rest after travelling a distance d metres from B. Find the value of d . [3]

8. A uniform rod AB of mass 8 kg and length 3 m rests horizontally in equilibrium supported at two points C and D, where AC = 0.5 m and AD = 2.5 m. A particle of mass 4 kg is placed on the rod at B. [10 marks]

- (a)** By taking moments about C, find the magnitude of the reaction force at D. [4]
- (b)** Hence find the magnitude of the reaction force at C. [2]

The particle of mass 4 kg is now moved to a new position X on the rod.

- (c)** Given that the reaction force at C is equal to the reaction force at D, find the distance AX. [4]

END OF PAPER